

HTML (HyperText Markup Language)

HTML is the standard markup language used to create and structure web pages. It provides the foundation for web development, enabling developers to design the skeleton of a webpage.

Key Features of HTML

1. **Structure:** Defines the layout and structure of web pages using elements.
 2. **Platform-Independent:** Works across all browsers and platforms.
 3. **Semantic Elements:** Introduced in HTML5, these elements improve SEO and accessibility.
 4. **Extensibility:** Works alongside CSS and JavaScript to build interactive and styled websites.
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Basic Concepts of HTML

1. Basic Structure of an HTML Document

```
<!DOCTYPE html>
<html>
<head>
  <title>My First Web Page</title>
</head>
<body>
  <h1>Welcome to My Website</h1>
  <p>This is a paragraph of text.</p>
</body>
</html>
```

Explanation:

- `<!DOCTYPE html>`: Specifies the HTML version (HTML5 in this case).
 - `<html>`: Root element that contains the entire document.
 - `<head>`: Contains metadata, like the title, character encoding, and linked resources.
 - `<title>`: Sets the title displayed on the browser tab.
 - `<body>`: Contains all visible content of the web page.
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2. HTML Tags

HTML uses **tags** to define elements. Tags typically come in pairs: an opening tag (`<tag>`) and a closing tag (`</tag>`). Some tags, like `<img>`, are self-closing.

Common Tags:

Tag	Description
<code><h1>-<h6></code>	Headings, from largest (<code>h1</code>) to smallest (<code>h6</code>).
<code><p></code>	Paragraph
<code><a></code>	Hyperlink
<code></code>	Image
<code></code> , <code></code> , <code></code>	Unordered and ordered lists, with list items.
<code><table></code>	Table structure
<code><div></code>	Generic container for styling or layout
<code></code>	Inline container for styling

3. Attributes

Attributes provide additional information about HTML elements and are specified in the opening tag.

Example:

```
<a href="https://www.example.com" target="_blank">Visit Example</a>
```

- `href`: Specifies the link's URL.
 - `target="_blank"`: Opens the link in a new tab.
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Intermediate HTML Concepts

1. Forms

Forms collect user input and send it to a server.

Example:

```
<form action="/submit" method="POST">
  <label for="name">Name:</label>
  <input type="text" id="name" name="name">
  <input type="submit" value="Submit">
</form>
```

Explanation:

- `<form>`: Defines the form. Attributes:
 - `action`: URL where the form data is sent.
 - `method`: HTTP method (`GET` or `POST`).
- `<input>`: Input field for user data.
- `<label>`: Label for form elements.

HTML provides various `<input>` tags for collecting user data in forms. Each `type` attribute of the `<input>` tag specifies the type of input expected from the user.

Basic Syntax

```
<input type="type" name="name" id="id" value="default_value"
placeholder="hint_text">
```

Types of Input Tags

Type	Description	Example
text	Single-line text input.	<code><input type="text" name="username" placeholder="Enter your name"></code>
password	Hidden characters for password input.	<code><input type="password" name="password"></code>
email	Email address input (validates format).	<code><input type="email" name="email" placeholder="example@mail.com"></code>
url	URL input (validates format).	<code><input type="url" name="website" placeholder="https://example.com"></code>

<code>number</code>	Numeric input (allows specifying <code>min</code> , <code>max</code> , <code>step</code>).	<code><input type="number" name="age" min="1" max="100" step="1"></code>
<code>tel</code>	Telephone number input (no validation, formats differ globally).	<code><input type="tel" name="phone" placeholder="123-456-7890"></code>
<code>search</code>	Search field with special styling.	<code><input type="search" name="search" placeholder="Search..."></code>
<code>date</code>	Date input (shows a date picker).	<code><input type="date" name="dob"></code>
<code>time</code>	Time input (shows a time picker).	<code><input type="time" name="appointment"></code>
<code>datetime-local</code>	Combined date and time input (local timezone).	<code><input type="datetime-local" name="schedule"></code>
<code>month</code>	Month input (YYYY-MM format).	<code><input type="month" name="month"></code>
<code>week</code>	Week input (Week-Year format).	<code><input type="week" name="week"></code>

color	Color picker for selecting colors.	<code><input type="color" name="favcolor"></code>
checkbox	Checkbox input for true/false values (can select multiple checkboxes).	<code><input type="checkbox" name="accept" value="yes"></code>
radio	Radio buttons for selecting one option from a group.	<code><input type="radio" name="gender" value="male"> Male</code>
file	File selection input (allows file uploads).	<code><input type="file" name="resume"></code>
submit	Submit button to send form data.	<code><input type="submit" value="Submit"></code>
reset	Reset button to clear form data.	<code><input type="reset" value="Reset"></code>
button	Generic button (used with JavaScript).	<code><input type="button" value="Click Me" onclick="alert('Hello')"></code>
image	Submit button in the form of an image.	<code><input type="image" src="submit.png" alt="Submit"></code>

<code>hidden</code>	Hidden field (used to pass data without user interaction).	<code><input type="hidden" name="userid" value="12345"></code>
<code>range</code>	Slider input for selecting a numeric value within a range.	<code><input type="range" name="volume" min="0" max="100"></code>
<code>datetime</code>	Deprecated: Use <code>datetime-local</code> instead.	-

Attributes for Input Tags

Attribute	Description	Example
<code>placeholder</code>	Displays hint text inside the input.	<code><input type="text" placeholder="Enter your name"></code>
<code>value</code>	Pre-fills the input field with a default value.	<code><input type="text" value="John Doe"></code>
<code>required</code>	Makes the field mandatory.	<code><input type="email" required></code>

<code>readonly</code>	Prevents user from modifying the input value.	<code><input type="text" value="Read-only" readonly></code>
<code>disabled</code>	Disables the input field (non-editable).	<code><input type="text" value="Disabled" disabled></code>
<code>maxlength</code>	Sets the maximum number of characters allowed.	<code><input type="text" maxlength="10"></code>
<code>min</code> and <code>max</code>	Defines the minimum and maximum values for numeric inputs.	<code><input type="number" min="1" max="10"></code>
<code>step</code>	Specifies the intervals for numeric inputs.	<code><input type="number" step="5"></code>
<code>size</code>	Sets the visible width of the input (number of characters).	<code><input type="text" size="20"></code>
<code>pattern</code>	Provides a regular expression to validate input.	<code><input type="text" pattern="[A-Za-z]+"></code>
<code>multiple</code>	Allows multiple values (e.g., file selection, email input).	<code><input type="file" multiple></code>

autofocus Automatically focuses on the input field when the page loads.

```
<input type="text" autofocus>
```

Examples for Input Types

Text Input

```
<form>
```

```
  <label for="name">Name:</label>
```

```
  <input type="text" id="name" name="name" placeholder="Enter your name" required>
```

```
</form>
```

Radio Button and Checkbox

```
<form>

  <label>

    <input type="radio" name="gender" value="male"> Male

  </label>

  <label>

    <input type="radio" name="gender" value="female"> Female

  </label>

  <br>

  <label>

    <input type="checkbox" name="subscribe" value="yes">
Subscribe to newsletter

  </label>

</form>
```

Range Input (Slider)

```
<form>

  <label for="volume">Volume:</label>

  <input type="range" id="volume" name="volume" min="0" max="100"
step="10">

</form>
```

File Upload

```
<form enctype="multipart/form-data">
```

```
<label for="file">Upload your file:</label>

<input type="file" id="file" name="file" accept=".pdf, .doc">

</form>
```

2. Tables

```
<table border="1">
  <tr>
    <th>First Name</th>
    <th>Last Name</th>
  </tr>
  <tr>
    <td>John</td>
    <td>Doe</td>
  </tr>
</table>
```

Explanation:

- `<table>`: Creates a table.
 - `<tr>`: Table row.
 - `<th>`: Table header.
 - `<td>`: Table cell.
-

3. Semantic Elements (HTML5)

Semantic elements describe the content's purpose, improving SEO and accessibility.

Element	Purpose
<header>	Defines the header section
<footer>	Defines the footer section
<article>	Represents an independent piece of content
<section>	Groups related content
<aside>	Represents content aside from the main content
<nav>	Defines navigation links

Example:

```
<header>
  <h1>My Website</h1>
  <nav>
    <a href="#home">Home</a>
    <a href="#about">About</a>
  </nav>
</header>
```

Advanced HTML Concepts

1. Multimedia

Embed audio and video in web pages.

Example: Audio

```
<audio controls>
```

```
<source src="song.mp3" type="audio/mpeg">
  Your browser does not support audio.
</audio>
```

Example: Video

```
<video controls width="500">
  <source src="movie.mp4" type="video/mp4">
  Your browser does not support video.
</video>
```

2. Canvas

The `<canvas>` element allows for dynamic, scriptable rendering of 2D and 3D graphics.

Example:

```
<canvas id="myCanvas" width="200" height="100" style="border:1px
solid #000;"></canvas>
<script>
  const canvas = document.getElementById("myCanvas");
  const ctx = canvas.getContext("2d");
  ctx.fillStyle = "red";
  ctx.fillRect(20, 20, 100, 50);
</script>
```

3. Responsive Design

Example: Meta Viewport

```
<meta name="viewport" content="width=device-width,
initial-scale=1.0">
```

Ensures the webpage is responsive on mobile devices.

Practice Problems

1. **Create a webpage with the following features:**
 - A navigation bar.
 - A form with fields for name, email, and a submit button.
 - A table displaying some data.
 - An embedded image and a video.
 2. **Create a webpage that uses semantic elements:**
 - Include a `<header>`, `<footer>`, `<article>`, and `<section>`.
 3. **Create a webpage with a canvas:**
 - Draw a rectangle, circle, and text on a canvas.
 4. **Create a responsive webpage:**
 - Include meta viewport, and style the page to adjust to different screen sizes.
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Interview Questions

1. **Basic Questions:**
 - What is HTML, and why is it important?
 - Explain the difference between block-level and inline elements.
 - What is the purpose of the `alt` attribute in images?
2. **Intermediate Questions:**
 - What are semantic elements? Why are they important?
 - How do forms work in HTML, and what are the different input types?
 - How do you create a table in HTML?
3. **Advanced Questions:**
 - What is the difference between `<div>` and `<span>`?
 - How do you embed multimedia (audio/video) in HTML?
 - Explain the `<canvas>` element and its use cases.
4. **Scenario-Based Questions:**

- How would you create a mobile-friendly webpage?
 - How would you optimize HTML for SEO?
 - How would you implement a simple user login form with validation?
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Real-World Applications

1. Building a Portfolio Website:

- Use HTML to structure the content and create pages like "About", "Projects", and "Contact."

2. Interactive Learning Platforms:

- Use HTML for quizzes and lessons with forms and multimedia content.

3. Online Store:

- Use tables and lists to display products, and forms for checkout.